Predicting Diabetes from Survey Data: An Analysis of the BRFSS 2015 Health Indicators

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I. Introduction

Diabetes is a common chronic condition in the United States and is often connected with other health problems, including high blood pressure, high cholesterol, higher body mass index (BMI), heart disease, and low physical activity. A diagnosis normally requires clinical testing, but not everyone has equal access to that type of care. Survey data such as the Behavioral Risk Factor Surveillance System (BRFSS) provides a less expensive way to collect information about people's health, behaviors, and demographics.

This makes the BRFSS data useful for an early screening problem. A model based only on survey responses would not replace a medical diagnosis, but it could help identify people who may benefit from additional testing. For that reason, we placed more importance on recall than precision during the classification portion of the project. Missing a likely diabetes case would be more concerning than recommending an additional test for someone who does not have diabetes.

A. Research Question

The main research question for this project is: *which self-reported health indicators, behaviors, and demographic characteristics are associated with diabetes status, and how accurately can those variables predict whether a respondent has diabetes?*

We approached this question in three parts. We first compared the characteristics of respondents with and without diabetes. We then used hypothesis tests and regression models to determine whether the observed differences were statistically significant. Finally, we trained several classification models and examined whether regularization, PCA, or clustering revealed additional structure in the data.

B. Hypotheses

Six hypotheses were tested. The first four focus on differences in BMI or associations with diabetes status. The final two examine whether physical activity and fruit consumption are related to BMI after including selected control variables.

H1 (t-test, BMI by diabetes status). H_0: there is no difference in mean BMI between individuals with diabetes and individuals without diabetes. H_a: there is a difference.

H2 (t-test, BMI by high blood pressure status). H_0: there is no difference in mean BMI between individuals with high blood pressure and individuals without it. H_a: there is a difference.

H3 (χ^2, diabetes and sex). H_0: diabetes status and sex are independent. H_a: they are associated.

H4 (χ^2, diabetes and stroke history). H_0: diabetes status and stroke history are independent. H_a: they are associated.

H5 (linear regression, physical activity and BMI). H_0: physical activity is not associated with BMI, controlling for sex and age. H_a: physical activity is negatively associated with BMI, controlling for sex and age.

H6 (linear regression, fruit consumption and BMI). H_0: daily fruit consumption is not associated with BMI, controlling for income and education. H_a: daily fruit consumption is negatively associated with BMI, controlling for income and education.

BMI was included in several hypotheses because it is closely related to diabetes risk. Physical activity and diet were also included because they are common targets of public health recommendations. H1 and H2 compare BMI across clinical groups, while H3 and H4 test whether diabetes is associated with sex and stroke history. H5 and H6 examine whether physical activity and fruit consumption remain associated with BMI after accounting for age, sex, income, and education.

In addition to the hypothesis tests, we developed a binary classifier for diabetes status (1 = yes, 0 = no) using eight selected indicators. We then compared regularized models and PCA-based models to determine whether either approach improved prediction.

II. Descriptive Statistics

A. Dataset

We used the `diabetes_binary_health_indicators_BRFSS2015` dataset from the CDC's 2015 BRFSS survey. The original dataset contains **253,680 rows** and **22 columns**. All columns were initially loaded as `float64`. Each row represents one survey respondent, and `Diabetes_binary` is the target variable.

The 21 predictor variables can be grouped into clinical, behavioral, and demographic or self-assessment measures. The clinical variables are `HighBP`, `HighChol`, `CholCheck`, `BMI`, `Stroke`, `HeartDiseaseorAttack`, and `DiffWalk`. Behavioral variables include `Smoker`, `PhysActivity`, `Fruits`, `Veggies`, and

TABLE I
DESCRIPTIVE STATISTICS FOR THE NUMERICAL VARIABLES
($n = 229{,}474$)

Statistic	BMI	PhysicalHealthDays
Count	229,474	229,474
Mean	28.69	4.68
Std. deviation	6.79	9.05
Minimum	12.00	0.00
25% (Q1)	24.00	0.00
50% (median)	27.00	0.00
75% (Q3)	32.00	4.00
Maximum	98.00	30.00

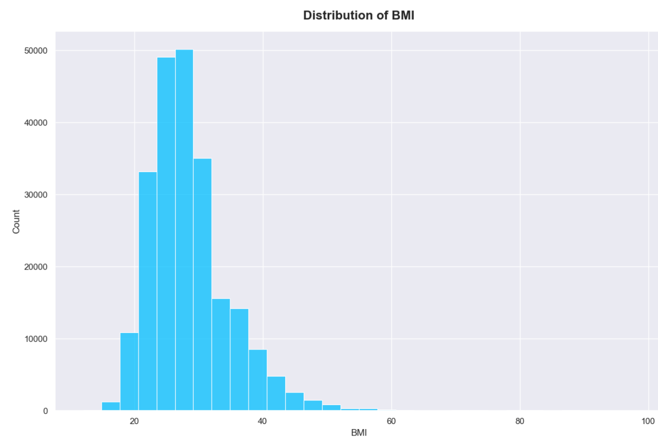

Fig. 1. Distribution of BMI. Most observations fall between 20 and 30, but the distribution has a long right tail extending to 98. The sample mean is in the Overweight category.

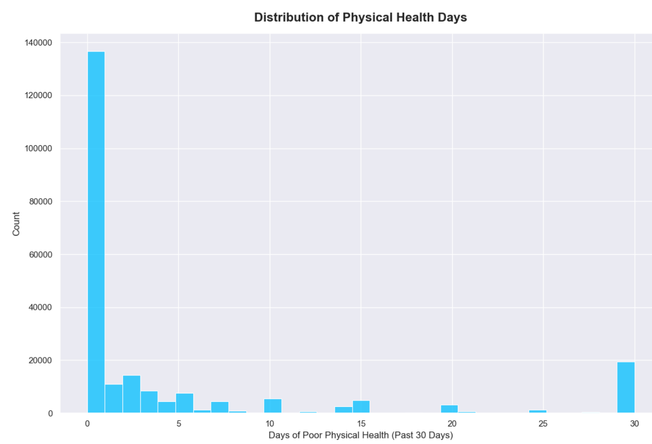

Fig. 2. Distribution of bad physical health days. A large share of respondents reported 0 days, while a smaller group reported values near 30. This produces a heavily right-skewed distribution.

`HvyAlcoholConsump`. The remaining variables describe health-care access, general health, mental and physical health days, sex, age, education, and income. Most variables are binary or ordinal. `BMI`, `MentHlth`, and `PhysHlth` are the main continuous or count-based variables.

B. *Preprocessing*

Type downcasting. The 22 columns initially used 42.6 MB because they were stored as `float64`. Since most columns only contain small integer codes, we converted 21 columns to `int8` and converted `BMI` to `float32`. Memory usage decreased to **6.0 MB** without changing the stored values.

Missing values. No missing values were found in any column, so imputation was not required.

Duplicates. We identified **24,206 duplicated rows**. These rows were removed to prevent identical response profiles from receiving extra weight during model training. The resulting dataset contained **229,474 rows**.

Renaming. Fifteen abbreviated columns were renamed to make the analysis easier to read. For example, `HighBP` became `HighBloodPressure`, and `GenHlth` became `GeneralHealth`.

Feature engineering. Eight labeled variables were created from the original survey codes. `GeneralHealthGroup` maps the values 1 through 5 to Excellent, Very Good, Good, Fair, and Poor. `BMIGroup` uses the CDC adult BMI categories. `MentalHealthDaysGroup` and `PhysicalHealthDaysGroup` divide the day counts into No Days, Some Days, and Frequent Days. We also created readable labels for sex, age, education, and income. Separate data frames for respondents with and without diabetes were created for group comparisons.

C. *Numerical Variables*

Table I summarizes the two numerical variables used most frequently in the analysis.

BMI. The average BMI is 28.69, compared with a median of 27.00, which indicates a right-skewed distribution. The middle 50% of observations falls between 24 and 32. The minimum is 12, while the maximum is 98, so the upper end contains several unusually large values. BMI was included because it is used directly in H1, H2, H5, and H6.

PhysicalHealthDays. This variable is also strongly right-skewed. The median is 0 even though the mean is 4.68, showing that many respondents reported no bad physical health days while a smaller group reported much larger values. The observed range is 0 to 30 days, and the middle 50% of the data falls between 0 and 4 days. The group comparisons later in the report show a noticeable difference between respondents with and without diabetes.

MentalHealthDays follows a similar pattern, with a mean of 3.51 and a median of 0.

The distributions are shown in Figures 1 and 2.

D. *Categorical Variables*

Table II reports the category counts for four variables used throughout the analysis.

The target variable is noticeably imbalanced. Approximately 84.7% of respondents do not have diabetes, while 15.3% do. As a result, a classifier that predicts the negative class for every respondent would achieve close to 85% accuracy without identifying any positive cases. This is why the later

TABLE II
CATEGORY COUNTS FOR KEY CATEGORICAL VARIABLES

Variable	Category	Count
Diabetes	0 (No diabetes)	194,377
	1 (Diabetes)	35,097
BMIGroup	Overweight	82,723
	Healthy Weight	58,842
	Class 1 Obesity	50,693
	Class 2 Obesity	20,448
	Class 3 Obesity	13,716
	Underweight	3,052
SexLabel	Female	128,715
	Male	100,759
AgeGroup	60-64	29,678
	65-69	29,093
	55-59	27,272
	50-54	23,121
	70-74	21,993
	45-49	17,280
	80+	16,791
	75-79	15,379
	40-44	14,040
	35-39	12,229
	30-34	10,023
	25-29	7,064
	18-24	5,511

TABLE III
MEAN BMI AND BAD HEALTH DAYS BY AGE GROUP

Age Group	Avg. BMI	Avg. Phys. Health Days	Avg. Mental Health Days
18-24	26.16	2.05	4.51
25-29	27.83	2.29	4.08
30-34	28.81	2.73	4.22
35-39	29.03	3.05	4.22
40-44	29.46	3.56	4.29
45-49	29.53	4.35	4.44
50-54	29.41	5.06	4.44
55-59	29.33	5.39	4.23
60-64	29.11	5.36	3.64
65-69	28.98	5.09	2.85
70-74	28.31	5.02	2.25
75-79	27.62	5.16	2.08
80+	26.21	5.71	1.83

model comparisons include precision, recall, and F1 rather than relying only on accuracy.

The `BMIGroup` counts are consistent with the BMI histogram. Overweight is the largest category, and the three obesity categories contain 84,857 respondents, or about 37% of the cleaned dataset. The sample contains more females than males, although both groups are well represented. The age distribution is concentrated more heavily in the older categories, especially ages 55 through 69. This is important because diabetes risk also changes with age.

E. Grouped Means

Two `groupby()` summaries were used to compare the numerical variables across age groups and diabetes status.

Table III shows that average BMI rises through early and middle adulthood, reaches its highest value of 29.53 for ages

TABLE IV
GROUP MEANS BY DIABETES STATUS

Measure	No Diabetes	Diabetes
Average BMI	28.10	31.96
Average Age (survey code)	7.85	9.38
Avg. bad physical health days	4.08	8.01
Avg. bad mental health days	3.33	4.49

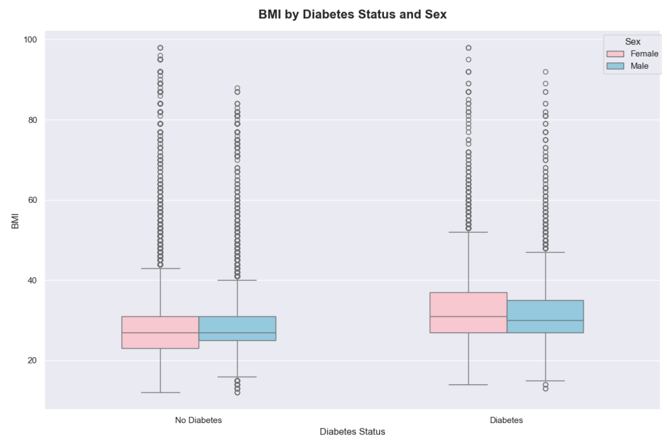

Fig. 3. BMI by diabetes status and sex. Median BMI is higher for both diabetic groups, while the differences between males and females within the same diabetes category are smaller.

45–49, and then declines in the older groups. Bad physical health days generally increase with age, from 2.05 among respondents ages 18–24 to 5.71 among respondents age 80 or older. Mental health days show a different pattern and decrease across the older age groups.

The differences by diabetes status are shown in Table IV. Respondents with diabetes have a higher average BMI, higher average age code, and more reported physical and mental health days. Average BMI increases from 28.10 for respondents without diabetes to 31.96 for those with diabetes. Bad physical health days increase from 4.08 to 8.01, which is nearly double.

F. Visualizations

Fourteen plots were created during EDA. Figures 3 through 6 show the visualizations that were most useful for the later analysis.

The remaining plots showed similar patterns. Median BMI increased as self-rated health became worse. Respondents with diabetes also had a wider distribution of bad physical health days. The single-variable bar plots showed diabetes rates of about 25% among respondents with high blood pressure, compared with about 7% among those without it. The rate was also higher among physically inactive respondents, smokers, and respondents with a stroke history. Heavy alcohol consumption was an exception, with a lower observed diabetes rate, although this should not be interpreted as a protective effect.

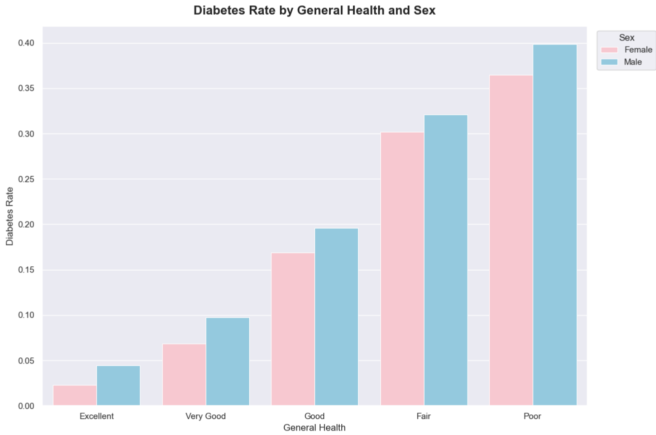

Fig. 4. Diabetes rate by general health and sex. Diabetes becomes much more common as self-rated health moves from Excellent to Poor. The differences between general-health categories are larger than the differences between males and females.

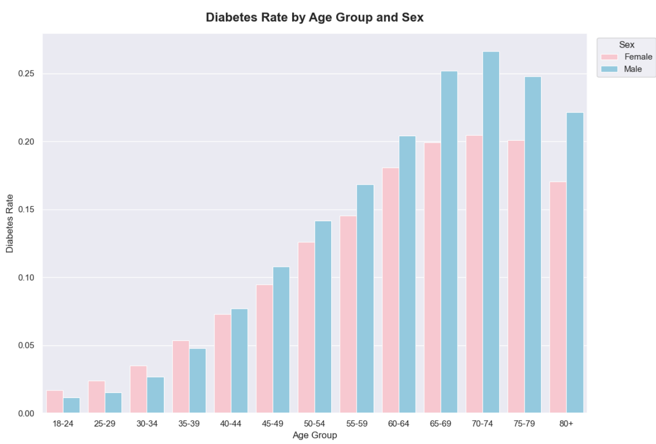

Fig. 5. Diabetes rate by age group and sex. Rates rise across most age groups, peak around ages 70–74, and then decrease slightly in the oldest groups.

The correlation heatmap was used as the custom visualization because it provided a compact view of the relationships among the variables and helped guide feature selection for classification.

III. Methodology

A. Data Wrangling and EDA

The data preparation steps followed the order described in Section II-B. We loaded the data, reduced the data types, checked for missing values, removed duplicates, renamed selected columns, and created the labeled variables used in the plots and grouped summaries. We also separated the data by diabetes status for several comparisons.

The EDA included descriptive statistics, category counts, two grouped summaries, and 14 visualizations. The cleaned data was saved as `data_cleaned.csv` and used as the input for the hypothesis testing, classification, PCA, and clustering stages. This kept the sample size consistent at 229,474 rows.

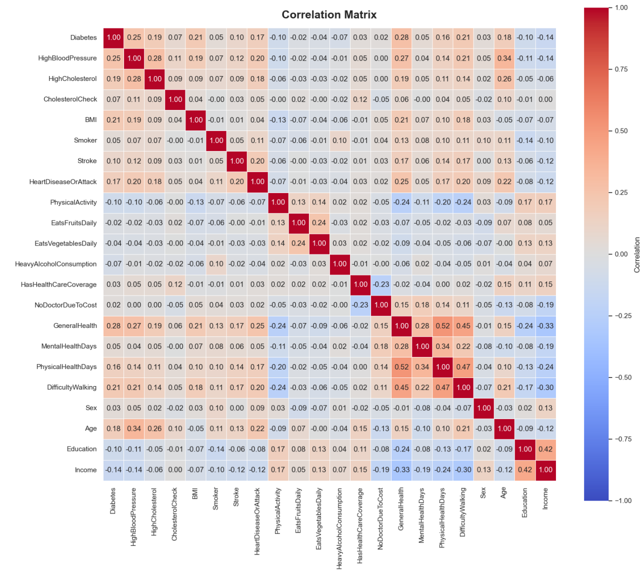

Fig. 6. Correlation matrix of the numeric variables. The largest positive correlations with diabetes are `GeneralHealth` (0.28), `HighBloodPressure` (0.25), `BMI` (0.21), and `DifficultyWalking` (0.21). No individual variable has a strong linear correlation with the target.

B. Hypothesis Testing

T-tests (H1, H2). Both tests use `BMI` as the dependent variable and were performed with `scipy.stats.ttest_ind`. H1 compares BMI by diabetes status, and H2 compares BMI by high blood pressure status. Both tests are two-tailed and use $\alpha = 0.05$.

BMI was right-skewed within each group rather than normally distributed. However, each group contained tens of thousands of observations. With samples of this size, the sampling distribution of the mean is expected to be close to normal, so we continued with the independent-samples t-tests.

Chi-square tests (H3, H4). The chi-square tests were performed with `scipy.stats.chi2_contingency` on 2×2 contingency tables created with `pd.crosstab`. H3 compares diabetes with sex, and H4 compares diabetes with stroke history. Both tests use one degree of freedom and $\alpha = 0.05$. All expected cell counts were well above the minimum needed for the chi-square approximation.

C. Linear Regression

Two ordinary least squares models were fit with `BMI` as the dependent variable. The main independent variable in each model is binary, so a scatter plot only produces two vertical groups of points. The regression coefficients were therefore used to estimate the direction and size of each association.

Model 1 (H5). The main predictor is `PhysicalActivity`, with `Sex` and `Age` included as controls. Age was entered using its ordinal survey code.

These controls were included because activity and BMI both vary across age and sex groups.

Model 2 (H6). The main predictor is `EatsFruitsDaily`. `IncomeGroup` and `EducationGroup` were converted into dummy variables using `pd.get_dummies(..., drop_first=True)`. The reference categories are income from $10,000 to $15,000 and College Graduate. Income and education were included because eating patterns may differ across socioeconomic groups.

D. Classification Setup

The classification task predicts `Diabetes`, where 1 represents diabetes and 0 represents no diabetes. Eight predictors were selected from the EDA and correlation results:

- `HighBloodPressure`, because it had one of the strongest correlations with diabetes.
- `HighCholesterol`, which is related to cardiovascular and metabolic health.
- `BMI`, which showed a clear difference by diabetes status.
- `HeartDiseaseOrAttack`, because it shares several risk factors with diabetes.
- `PhysicalActivity`, which represents a modifiable health behavior.
- `GeneralHealth`, which had the largest correlation with the target.
- `Sex`, which was associated with diabetes in H3.
- `AgeGroup`, because diabetes rates increased across most age groups.

`AgeGroup` was one-hot encoded using `OneHotEncoder(sparse_output=False, drop=None, handle_unknown='ignore')`. This produced 13 age columns. Because no age category was dropped, the dummy variables contain one exact linear dependency. Together with the seven remaining predictors, the final feature matrix had **20 columns**.

The data was divided into training and test sets with `train_test_split(test_size=0.2, random_state=42)`. The training set contained 183,579 rows, and the test set contained 45,895 rows.

Balancing. The training set contained 155,564 negative cases and 28,015 positive cases. We used random undersampling to reduce the negative class to 28,015 observations, producing a balanced training set of 56,030 rows. Undersampling was selected because the original dataset was large enough to retain a substantial training sample after removing majority-class rows.

Only the training set was balanced. The test set kept the original class distribution so that the final metrics would reflect the full dataset more closely. Training on a 50/50 sample and testing on a dataset with only about 15% positive cases was expected to lower precision. We accepted this tradeoff because recall was the main screening metric.

The features were not standardized for this part of the analysis. This choice mainly affected SVC, as discussed in the results.

E. Model Selection and Regularization

Four classifiers were compared with 5-fold cross-validation on the balanced training set. We used `KFold(n_splits=5, shuffle=True, random_state=42)` with `cross_validate` and recorded accuracy, precision, recall, and F1.

- `SVC(max_iter=10000)`
- `DecisionTreeClassifier(max_depth=10, random_state=42)`
- `RandomForestClassifier(n_estimators=100, random_state=42)`
- `GradientBoostingClassifier(n_estimators=100, learning_rate=0.1)`

The classifier with the highest cross-validated F1 score was fit again on the full balanced training set and evaluated on the test set.

Three regularized logistic regression models were also trained on the same balanced data and evaluated on the same test set:

- **L1**: `solver='liblinear'`, `penalty='l1'`, `C=0.1`. L1 can push coefficients to exactly zero, so it does implicit feature selection.
- **L2**: `penalty='l2'`, `C=1.0`. L2 shrinks coefficients without eliminating them.
- **Elastic Net**: `solver='saga'`, `penalty='elasticnet'`, `C=0.001`, `l1_ratio=0.5`, an even mix of both.

The Elastic Net model used a much smaller value of C than the L1 and L2 models. A stronger penalty was needed to produce a noticeable number of zero coefficients. Because of this, the Elastic Net results reflect both the type of penalty and a difference in penalty strength.

F. PCA and K-Means

PCA. The 20-column feature matrix was standardized with `StandardScaler` before PCA. Standardization was necessary because BMI has a much wider numerical range than the binary and dummy variables. PCA was first fit with all components, and we selected the smallest number of components needed to exceed 90% cumulative explained variance.

The PCA-transformed data was split 80/20 using the same random seed. A default `LogisticRegression` model was evaluated with 5-fold cross-validation and on the test set using accuracy and F1. Unlike the earlier classification analysis, this data was not balanced.

K-Means. K-Means was run on the standardized feature matrix for $k = 1$ through $k = 20$. Inertia was recorded for each value, and the final number of clusters was selected from the elbow plot. The fitted cluster centers were converted to a labeled DataFrame and interpreted in standardized units. Diabetes status was not included as an input to clustering.

G. Summary of Decisions

The main preprocessing choices were kept consistent across the project. Duplicates were removed, no imputation was

TABLE V
TWO-SAMPLE T-TEST RESULTS (BMI AS DEPENDENT VARIABLE)

	Grouping Variable	t	p-value
H1	Diabetes status	100.38	0.0
H2	High blood pressure	94.84	0.0

TABLE VI
CONTINGENCY TABLES AND CHI-SQUARE RESULTS (DF = 1)

H3: Diabetes by Sex		
	Female (0)	**Male (1)**
No diabetes (0)	110,370	84,007
Diabetes (1)	18,345	16,752
$\chi^2 = 245.55$, $p = 2.42 \times 10^{-55}$		
H4: Diabetes by Stroke History		
	No stroke (0)	**Stroke (1)**
No diabetes (0)	187,361	7,016
Diabetes (1)	31,829	3,268
$\chi^2 = 2256.53$, $p = 0.0$		

needed, and the right-skewed BMI values were left unchanged. Age was one-hot encoded without dropping a category for classification and PCA, while income and education used a reference category in the regression model. The classification training data was undersampled, and standardization was applied before PCA and K-Means but not before the initial classifier comparison.

IV. RESULTS

A. T-tests

Both t-test p-values were displayed as 0.0 because they were below the floating-point precision shown in the output. In both cases, the values were far below 0.001. We therefore reject both null hypotheses at $\alpha = 0.05$.

H1. Mean BMI differs significantly between respondents with and without diabetes. The result agrees with the group means in Table IV, where the averages are 31.96 and 28.10, and with Figure 3.

H2. Mean BMI also differs significantly between respondents with and without high blood pressure.

B. Chi-Square Tests

H3. The test produced $\chi^2 = 245.55$ with one degree of freedom and $p = 2.42 \times 10^{-55}$. We reject H_0 and conclude that sex and diabetes status are associated in this sample.

H4. The stroke-history test produced $\chi^2 = 2256.53$ and a p-value reported as 0.0. We again reject H_0. The chi-square statistic is much larger than the statistic for sex, which indicates a stronger departure from independence. This is consistent with the observed diabetes rates of 31.8% among respondents with a stroke history and 14.5% among those without one.

C. Linear Regression

Model 1 (H5). The coefficient for `PhysicalActivity` is -2.059 with a p-value reported as 0.0. We reject H_0.

TABLE VII
MODEL 1, OLS COEFFICIENTS (DV: BMI). $R^2 = 0.02113$, ADJ. $R^2 = 0.02112$

Term	Coef.	SE	t	p
Intercept	31.059	0.049	633.04	0.0
PhysicalActivity	-2.059	0.032	-64.68	0.0
Sex	0.458	0.028	16.20	5.58×10^{-59}
Age	-0.132	0.005	-28.90	2.53×10^{-183}

Holding sex and age constant, physically active respondents have an average BMI about 2.06 units lower than inactive respondents. The coefficients for sex and age are also statistically significant. Males have an estimated BMI 0.46 units higher than females, while BMI decreases by about 0.13 units for each increase in the age survey code.

The model has an R^2 of only **0.02113**, meaning that the three predictors explain about 2.1% of the variation in BMI. The coefficients are statistically significant, but the overall explanatory power is weak. The large sample size makes it possible for relatively small effects to produce very small p-values.

Model 2 (H6). The coefficient for `EatsFruitsDaily` is -0.839 with $p = 6.01 \times 10^{-183}$. We reject H_0. Respondents who reported eating fruit daily have an estimated BMI 0.84 units lower than respondents who did not, after controlling for income and education.

Most income coefficients are negative relative to the $10,000 to $15,000 reference group, which indicates lower estimated BMI in the higher income categories. The Less than $10,000 category is not statistically different from the reference group. Every education category below College Graduate has a positive coefficient, ranging from 0.80 to 1.31, and each is statistically significant at $\alpha = 0.05$.

The second model has an R^2 of **0.012376**, so it explains about 1.2% of the variation in BMI. As with Model 1, the statistical associations are clear even though the model explains little of the total variation.

D. Model Selection

Gradient Boosting produced the best overall cross-validation results, with 73.3% accuracy, 71.7% precision, 77.0% recall, and a 74.3% F1 score. It had the highest accuracy, precision, and F1, while its recall was also close to the highest value. Based on F1, it was selected for test-set evaluation.

SVC achieved 49.9% accuracy and 94.9% recall, which indicates that it predicted the positive class for most observations. The lack of feature scaling likely contributed to this result because BMI has a much wider range than the binary predictors. The model also produced convergence warnings. Random Forest performed below the single Decision Tree in this comparison.

E. Test Performance and Regularization

Gradient Boosting achieved 70.8% accuracy, 31.6% precision, 76.8% recall, and a 44.8% F1 score on the test

TABLE VIII

MODEL 2, OLS COEFFICIENTS (DV: BMI). REFERENCE CATEGORIES: INCOME $10,000 TO $15,000, EDUCATION COLLEGE GRADUATE. $R^2 = 0.012376$, ADJ. $R^2 = 0.01232$

Term	Coef.	SE	t	p
Intercept	29.485	0.071	415.91	0.0
EatsFruitsDaily	-0.839	0.029	-28.87	6.01×10^{-183}
Income: $15,000 to <$20,000	-0.492	0.082	-5.99	2.13×10^{-9}
Income: $20,000 to <$25,000	-0.685	0.079	-8.71	2.99×10^{-18}
Income: $25,000 to <$35,000	-0.791	0.076	-10.44	1.59×10^{-25}
Income: $35,000 to <$50,000	-0.764	0.073	-10.48	1.06×10^{-25}
Income: $50,000 to <$75,000	-0.790	0.072	-10.94	7.37×10^{-28}
Income: $75,000+	-1.160	0.070	-16.65	3.30×10^{-62}
Income: Less than $10,000	-0.100	0.092	-1.08	0.281 (n.s.)
Education: Never Attended School or Kindergarten Only	1.305	0.513	2.55	1.09×10^{-2}
Education: Completed Grades 1 through 8	0.880	0.112	7.87	3.45×10^{-15}
Education: Completed Grades 9 through 11	1.103	0.076	14.42	3.87×10^{-47}
Education: High School Graduate or GED	0.801	0.038	21.09	1.31×10^{-98}
Education: Some College or Technical School	0.838	0.036	23.45	2.05×10^{-121}

TABLE IX

5-FOLD CROSS-VALIDATION ON THE BALANCED TRAINING SET

Model	Acc.	Prec.	Recall	F1
SVC	0.4990	0.4997	**0.9495**	0.6546
Decision Tree	0.7183	0.7015	0.7601	0.7295
Random Forest	0.6874	0.6818	0.7029	0.6921
Gradient Boosting	**0.7334**	**0.7173**	0.7705	**0.7429**

TABLE X

TEST SET PERFORMANCE (45,895 ROWS, NATURAL 85/15 SPLIT)

Model	Acc.	Prec.	Recall	F1
Gradient Boosting	0.7078	0.3161	**0.7676**	0.4478
Logistic + L1	**0.7157**	**0.3206**	0.7528	**0.4497**
Logistic + L2	0.7153	0.3203	0.7533	0.4495
Logistic + Elastic Net	0.7131	0.3132	0.7207	0.4367

set. Recall remained close to the cross-validation result, but precision dropped substantially. Since the model was trained on a balanced sample and evaluated on the original imbalanced test set, it predicted diabetes more often than the natural class distribution would suggest.

The L1 model produced 71.57% accuracy, 32.06% precision, 75.28% recall, and a 44.97% F1 score. These results are nearly the same as Gradient Boosting. L1 set only `AgeGroup_55-59` exactly to zero. `PhysicalActivity` was also reduced to a small coefficient of -0.044, while `GeneralHealth` and `HighBloodPressure` retained larger coefficients.

The L2 model achieved 71.53% accuracy, 32.03% precision, 75.33% recall, and a 44.95% F1 score. It retained all 20 coefficients, as expected from L2 regularization. Its performance was almost identical to L1.

Elastic Net achieved 71.31% accuracy, 31.32% precision, 72.07% recall, and a 43.67% F1 score. It set 12 of the 20 coefficients to zero, including `PhysicalActivity` and most age-group variables. The remaining coefficients were smaller than those from the L1 and L2 models.

TABLE XI

REGULARIZED LOGISTIC REGRESSION COEFFICIENTS

Feature	L1	L2	Elastic Net
HighBloodPressure	0.736	0.732	0.711
HighCholesterol	0.530	0.530	0.462
BMI	0.068	0.069	0.059
HeartDiseaseOrAttack	0.294	0.294	0.147
PhysicalActivity	-0.044	-0.043	**0.000**
GeneralHealth	0.561	0.563	0.528
Sex	0.209	0.215	0.018
Age 18-24	-1.456	-1.647	**0.000**
Age 25-29	-1.417	-1.547	**0.000**
Age 30-34	-1.227	-1.313	**0.000**
Age 35-39	-0.785	-0.847	**0.000**
Age 40-44	-0.519	-0.574	**0.000**
Age 45-49	-0.307	-0.354	**0.000**
Age 50-54	-0.135	-0.176	**0.000**
Age 55-59	**0.000**	-0.023	**0.000**
Age 60-64	0.232	0.208	**0.000**
Age 65-69	0.445	0.422	0.062
Age 70-74	0.538	0.518	0.065
Age 75-79	0.415	0.399	**0.000**
Age 80+	0.380	0.365	**0.000**

The test F1 scores were 44.97% for L1, 44.95% for L2, 44.78% for Gradient Boosting, and 43.67% for Elastic Net. The difference between the highest and lowest value was only 0.013. Elastic Net gave up the most recall, which is a disadvantage for the screening goal of this project.

Overall, regularization changed the coefficient patterns much more than it changed predictive performance. Elastic Net removed more than half of the features while producing an F1 score close to the other models. This suggests that much of the useful signal is concentrated in a smaller group of variables.

F. PCA

PC1 explains **10.5% of the total variance**. Components 2 through 14 each explain approximately 5% to 6%. Reaching 90% cumulative explained variance requires **16 of the 20 com-**

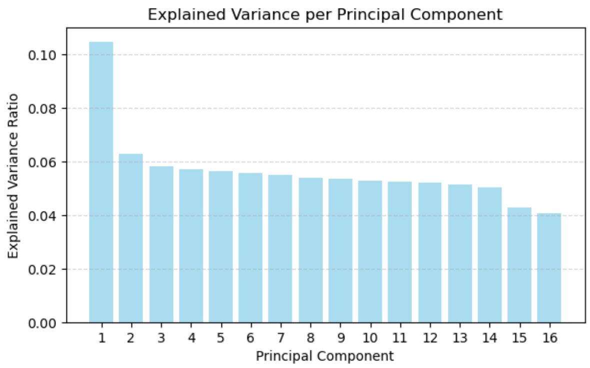

Fig. 7. Explained variance ratio for the first 16 principal components. PC1 accounts for the largest share, but the remaining bars are relatively similar in height.

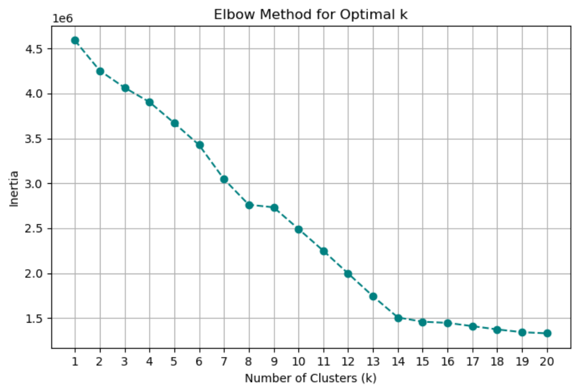

Fig. 8. Elbow plot for K-Means. The decrease in inertia becomes noticeably smaller after approximately $k = 8$, which was selected for the final model.

ponents, with a cumulative value of 0.9011. PCA therefore provides only a small reduction in the number of features.

The 20th component explains 0.0000 variance. This occurs because the 13 one-hot-encoded age columns sum to 1 for every observation, creating one exact linear dependency in the feature matrix.

A logistic regression model using the 16 PCA components achieved 85.13% cross-validated accuracy and a 0.2159 F1 score. On the test set, it achieved 84.99% accuracy and a 0.2174 F1 score. Because this stage used the original imbalanced classes, the model favored the negative class. Its high accuracy is close to the 84.7% negative-class rate, while its F1 score is much lower than the scores from the balanced models.

G. K-Means Clustering

The elbow plot suggested $k = 8$. The cluster centers in Table XII show that age is the main variable separating the clusters. The clusters can therefore be read as health profiles that correspond mostly to different age ranges.

TABLE XII
K-MEANS CLUSTER PROFILES ($k = 8$), STANDARDIZED UNITS

Cl.	Dominant Age	HighBP	HighChol	BMI
3	18-34, 40-44	−0.547	−0.485	−0.032
0	35-39	−0.497	−0.419	0.051
2	45-49	−0.267	−0.169	**0.124**
4	50-54	−0.122	−0.050	0.106
5	55-59	0.017	0.073	0.095
6	60-64	0.144	0.167	0.062
7	70-74	**0.359**	**0.280**	−0.055
1	65-69, 75-79, 80+	0.342	0.227	−0.119

Cluster 3 contains most of the youngest respondents and has the lowest standardized values for high blood pressure, high cholesterol, and heart disease, along with the highest physical activity. Clusters 0, 2, 4, 5, and 6 move through the middle age ranges, and the chronic-condition measures generally increase across those clusters. Clusters 7 and 1 contain the oldest respondents and have the highest values for high blood pressure and high cholesterol.

BMI does not increase in the same way. It reaches its highest standardized value in cluster 2, which represents middle-aged respondents, and falls below the sample average in the two oldest clusters. Sex remains close to the sample average in every cluster, so it contributed little to the separation. Overall, the clusters mainly describe increasing age, lower activity, and a higher prevalence of chronic health conditions.

V. CONCLUSION

A. Key Takeaways

All six hypotheses were supported by the statistical results. The t-tests and chi-square tests were significant at $\alpha = 0.05$, and both regression models found a negative association between the main behavioral predictor and BMI. Respondents with diabetes were older on average, had a higher average BMI, and reported more bad physical health days.

The regression results also showed why statistical significance should not be considered by itself. Both models produced extremely small p-values, but their R^2 values were only 0.021 and 0.012. With more than 229,000 observations, small effects can be statistically significant even when the model explains very little of the variation in BMI.

Class imbalance had a large effect on model evaluation. The PCA logistic regression achieved about 85% accuracy but an F1 score near 0.22 because it favored the majority class. The models trained on undersampled data had lower accuracy but F1 scores near 0.45 and much higher recall for diabetes cases.

The final model results were also very similar. Gradient Boosting, L1, L2, and Elastic Net were separated by only 0.013 in test F1. Elastic Net removed 12 of the 20 coefficients with only a small loss in performance, indicating that several predictors added little unique information.

PCA did not reveal a low-dimensional representation of the data. Sixteen of the 20 components were needed to explain 90% of the variance, and the first component explained only 10.5%. The selected predictors appear to provide several separate sources of weak information rather than a small number of dominant underlying factors.

The clinical and self-rated health variables were the most consistent predictors. `GeneralHealth`, `HighBloodPressure`, `HighCholesterol`, and `HeartDiseaseOrAttack` retained larger coefficients across the regularized models. `PhysicalActivity` was reduced to zero by Elastic Net. The K-Means results also showed that age was the main source of separation among the clusters.

B. Limitations

The dataset has several limitations. All measures are self-reported, so the responses may contain recall error or social desirability bias. BMI is based on reported height and weight, and `GeneralHealth`, the strongest single correlate of diabetes, is a subjective five-point rating.

The data is cross-sectional, which means the results show associations but not cause-and-effect relationships. For example, the analysis cannot determine whether low physical activity contributes to higher BMI or whether a higher BMI makes physical activity more difficult. The lower observed diabetes rate among heavy drinkers is also likely influenced by other variables and should not be interpreted as a health benefit.

Removing the 24,206 duplicate rows was another limitation. Since the dataset does not include a respondent identifier and many variables use broad categories, some identical rows may represent different people. Random undersampling also removed roughly 127,000 negative training observations, so some potentially useful information was discarded.

One-hot encoding every age category introduced perfect collinearity and created the zero-variance final principal component. Most models were also run with default or near-default hyperparameters. The Part 3 features were not scaled, which affected SVC, and the Elastic Net model used a stronger penalty than the L1 and L2 models. This makes the regularization comparison less controlled than it could have been.

C. Future Work

A first improvement would be to standardize the predictors before fitting the Part 3 models so that SVC can be evaluated under more appropriate conditions. Hyperparameter tuning for Gradient Boosting and the other classifiers would also help determine whether performance can improve beyond the current results. The three regularization methods should be compared over the same grid of C values.

Future work should also compare random undersampling with SMOTE, class-weighted models, and decision-threshold tuning. Threshold tuning may be especially useful because it can change the balance between precision and recall without removing training observations.

The classification models used only 8 of the 21 available predictors. Adding `DifficultyWalking`, `Income`, `Education`, and the mental and physical health-day variables may improve performance. Interaction terms could also be tested, especially for age and sex, since Figure 5 suggests that their relationship with diabetes changes across age groups.